MPPI Path Following with Foothold-Quality-Aware Cost for Quadruped Navigation in Unstructured Terrain*

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***Abstract*— Local planners used in mobile robot navigation evaluate trajectories based on center-of-mass collision checks, sufficient for wheeled platforms but inadequate for legged robots, whose stability depends on discrete foothold quality. We present a hierarchical framework that addresses this by augmenting a Model Predictive Path Integral (MPPI) local planner with a *Foothold-Quality-Aware (FQA) critic*, which penalizes trajectories over unsteppable terrain, and a *Turn-in-Place critic*, which suppresses destabilizing lateral motion. These are integrated with terrain-aware A* global planning and a convex centroidal MPC enhanced with slope-aligned friction cones, COM adaptation, and capture-point stepping. Ablation studies on a simulated Unitree Go2 in dense forest environments show that the full system achieves stable, fall-free traversal, while removing any component degrades traversal stability.**

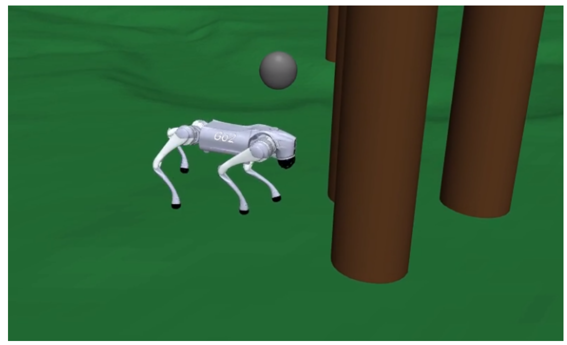

Fig. 1. The Unitree Go2 quadruped in a British Columbia forest pointcloud reconstructed on MuJoCo

I. Introduction

Autonomous quadruped robots possess an inherent structural advantage over wheeled vehicles in navigating unstructured, hazardous, and uneven environments. By decoupling the body from the ground via articulated legs, quadrupeds can actively control their base posture, step over obstacles, and traverse discontinuous terrain such as forest floors, disaster sites, and rubble [1], [2]. However, unlocking this mobility requires a navigation stack capable of reasoning about the environment at multiple scales: from long-horizon global path planning down to high-frequency, contact-rich stabilizing leg control [3].

The standard paradigm for autonomous navigation employs a hierarchical architecture. A global planner (e.g., A* or D* Lite) generates a collision-free path across a static grid, while a local planner such as the Dynamic Window Approach (DWA) or Model Predictive Path Integral (MPPI) control [4], [5] computes reactive body-frame velocity commands to track the path and avoid unmapped dynamic obstacles. While heavily optimized for wheeled platforms (e.g., the ROS 2 Nav2 stack [6]), directly applying these local planners to legged robots introduces a significant gap: standard trajectory rollouts evaluate collisions and costs strictly based on the robot's Center of Mass (COM) footprint.

For a wheeled robot, clearing the COM footprint is physically sufficient. For a quadruped, it is fundamentally inadequate. A quadruped's support polygon extends well beyond its body footprint. If a local planner commands the robot along a path where the COM is collision-free, but the required footholds lie on steep slopes, ridges, or deep gaps, the low-level locomotion controller will struggle to maintain contact stability, resulting in foot slippage, kinematic saturation, or catastrophic falls [7].

Most contemporary solutions address this entirely at the low-level controller by employing terrain-aware heuristic foothold selection [8] or learning-based foothold adaptations [9], [10]. However, placing the burden of terrain navigation solely on the reactive leg controller is suboptimal; if the local planner steers the robot into a region devoid of viable footholds, no amount of reactive adaptation can recover stability. A true solution requires the local path planner to be inherently *locomotion-aware*. In this paper, we propose a novel formulation of the MPPI local planner specifically designed for quadruped robots navigating unstructured forest-like terrain simulated on MuJoCo. Rather than evaluating costmaps via a monolithic COM footprint, we introduce a Foothold-Quality-Aware (FQA) critic that explicitly predicts the kinematics of all four legs along simulated rollouts, penalizing trajectories that require stepping on highly uneven or sloped surfaces. Furthermore, to address the destabilizing "crab-walking" (simultaneous forward and lateral motion) often observed when quadruped base velocities are commanded by omnidirectional planners, we introduce a Turn-in-Place critic that enforces stability-preserving heading alignment.

The specific contributions of this work are threefold:

1) We introduce an FQA cost function for the MPPI planner that evaluates anticipated ground contact locations against a continuously derived terrain steppability map.

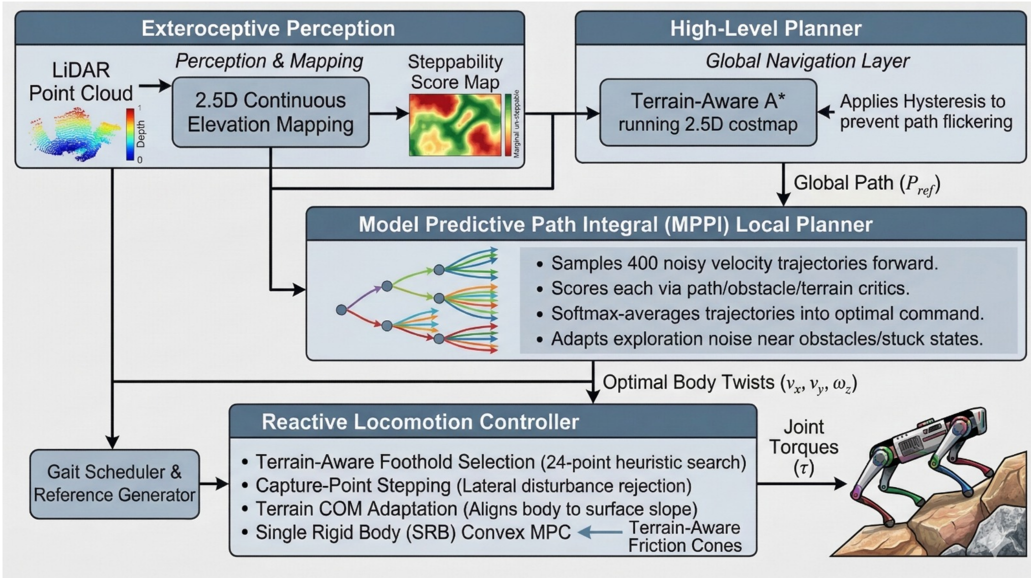

Fig. 2. System architecture of the quadruped navigation framework.

2) We present a customized MPPI multi-critic architecture that actively suppresses unstable lateral quadruped kinematics by heavily penalizing large heading errors.
3) We integrate these contributions into a complete, real-time in simulation hierarchical framework—combining perception, terrain-aware A*, MPPI, and a convex centroidal Model Predictive Controller (MPC)—and demonstrate its effectiveness in dense, unstructured simulated forest environments.

The remainder of this paper is organized as follows. Section II reviews related work in quadruped navigation and sampling-based planning. Section III explicitly details the hierarchical architecture, the derivation of the FQA critic, and the low-level MPC tracking. Section IV presents quantitative navigation metrics and qualitative behavior analysis, followed by ablation studies isolating the impact of our proposed critics in Section V. Section VI concludes the paper.

II. Related Work

Model Predictive Control (MPC) has become the standard for highly dynamic quadruped locomotion, functioning effectively across varied terrain. Typically, these formulations employ Convex MPC based on simplified centroidal dynamics [11] or tackle the full Whole-Body Nonlinear MPC [12] to account for explicit contact dynamics and kinematic constraints. While gradient-based methods excel at stabilizing continuous nonlinear dynamics, they struggle with severe non-convexities, such as discrete obstacle fields and discontinuous cost landscapes characteristic of rough terrain.

Recently, Model Predictive Path Integral (MPPI) control [5] and generalized sampling-based MPC have gained traction in legged robotics to bypass gradient computation hurdles. Recent studies have demonstrated the deployment of whole-body sampling-based MPC on hardware [13], highlighting its capacity to navigate complex local minima without pre-defined contact schedules. Efforts to improve the sample efficiency of MPPI in locomotion have explored varied sampling distributions, such as spline-based action sequences [14] and integration with Covariance Matrix Adaptation (CMA-ES) in the MPOPI framework [15]. However, running MPPI at the low-level torque or contact force level remains computationally demanding for extended spatial horizons. In contrast to these whole-body sampling planners, our framework utilizes MPPI at the *local navigation tier* operating in $SE(2)$, relying on standard Convex MPC for the low-level stabilization. This hierarchical split allows the MPPI planner to evaluate extensive spatial rollouts using terrain costmaps at high frequency.

To effectively utilize such terrain costmaps, selecting secure footholds is paramount for mitigating slip and ensuring stability on irregular topographies. Early frameworks combined geometric heuristics with search trees (e.g., A* or RRT*) to find kinematic plans over challenging obstacles, extensively demonstrated on platforms like LittleDog [16] and HyQ [17]. More recent methods quantify terrain desirability using dense 2.5D elevation maps. Fankhauser et al. [3] demonstrated autonomous stair climbing on ANYmal using continuous elevation mapping, extracting surface normals and roughness to heuristically score footholds. Similarly, Mastalli et al. [8] jointly optimized COM trajectories and footholds over terrain costmaps using CMA-ES, and Li et al. [18] utilized Support Vector Machines (SVM) to learn foothold cost functions directly from expert demonstrations. Our work synergizes terrain evaluation directly into the continuous path generation layer. Rather than decoupling the path optimization from discrete footstep planning, our Foothold-Quality-Aware (FQA) critic continuously evaluates proposed SE(2) base trajectories by probabilistically sampling the underlying terrain quality where the robot's feet are predicted to land.

This integration of terrain-aware planning requires a robust system design. State-of-the-art quadruped stacks inherently rely on multi-layered architectures, decomposing the problem into global mapping, local trajectory planning, and high-frequency centroidal/whole-body control [1], [2]. For autonomous navigation, these stacks frequently integrate variations of graph searches for global routes coupled with reactive local planners. In the broader mobile robotics domain, architectures like the ROS 2 Navigation stack (Nav2) employ MPPI controllers as well as local planners (e.g., DWB, TEB) that score velocity rollouts using footprint-based or 3D bounding box collision cost functions. Unfortunately, applying standard wheeled-robot critics to legged platforms drastically limits their mobility, as quadrupeds can actively step over obstacles that would obstruct a wheeled chassis. Our approach bridges the gap between conventional wheeled navigation and legged mobility by reformulating the MPPI local planner cost function. By scoring predicted discrete leg placements rather than treating the robot as a rigid footprint, the planner permits traversal over sparsely obstructed terrain. Furthermore, our architecture incorporates a turn-in-place (yaw) critic specifically tuned for quadrupeds, which penalizes lateral and diagonal velocity commands that risk displacing the center of mass into dynamically unstable configurations — reducing the likelihood of the characteristic "crab-walk" singularities commonly induced by such commands in standard MPC tracking modules.

III. Proposed Methodology

A. System Overview

The proposed framework decomposes autonomous quadruped navigation into three layers operating at distinct spatial and temporal scales (Fig. 2):

1) **Global planning** (III-C): A terrain-aware A* planner computes a collision-free waypoint path on a 2D cost grid at ∼12 Hz.
2) **Local trajectory optimization** (III-D): An MPPI controller generates body-frame velocity commands (v_x, v_y, ω_z) that track the global path while satisfying legged-locomotion-specific costs, at ∼24 Hz.
3) **Whole-body control** (III-E): A convex centroidal MPC resolves optimal ground reaction forces at ∼48 Hz; a leg controller maps these to joint torques at 200 Hz.

The key insight motivating this hierarchy is that global planners reason over large spatial extents but use simplified collision models, while low-level controllers handle full rigid-body dynamics but have no notion of terrain quality beyond the immediate contact patch. The MPPI layer bridges this gap by evaluating candidate trajectories against *leg-specific* terrain costs; the core contribution of this work.

B. Perception and Terrain Representation

1) Elevation Mapping: A simulated 3D LiDAR ($N_{\mathrm{az}} \times N_{\mathrm{el}} = 90 \times 15 = 1350$ rays, $d_{\max} = 6$ m) generates point clouds that are accumulated into a fixed-frame elevation grid $\mathscr{G}$ of extent $W \times W = 12$ m $\times$ 12 m at resolution $\delta = 0.05$ m. Each cell (i, j) maintains two height estimates updated via exponential moving average (EMA):

$$h_g^{(i,j)} \leftarrow (1-\alpha_g)\, h_g^{(i,j)} + \alpha_g \cdot Q_{q_g}\left(\{z_k\}_{k\in(i,j)}\right)$$

$$h_t^{(i,j)} \leftarrow (1-\alpha_t)\, h_t^{(i,j)} + \alpha_t \cdot \max\left(\{z_k\}_{k\in(i,j)}\right)$$

where $\alpha_g = 0.25$, $\alpha_t = 0.50$, h_g is the ground layer ($q_g = 0.20$ quantile of observed heights, providing a robust walkable-surface estimate), and h_t is the top layer (maximum observed height, capturing obstacle surfaces). The per-cell **clearance** $c^{(i,j)} = h_t^{(i,j)} - h_g^{(i,j)}$ distinguishes terrain variation from protruding obstacles.

2) Steppability Score: Drawing inspiration from continuous elevation mapping heuristics [3], [8], we define a continuous terrain quality metric $\sigma : \mathbb{R}^2 \to [0,1]$ that combines two complementary indicators of foothold viability where the first term accounts for slope and the second for roughness:

$$\sigma(x,y) = w_s \cdot \min\left(\frac{\|\nabla h_g(x,y)\|}{s_{\max}}, 1\right) + w_r \cdot \min\left(\frac{\mathrm{std}\left(h_g^{\mathscr{N}}\right)}{r_{\max}}, 1\right) \tag{1}$$

with $w_s = 0.6$, $w_r = 0.4$, $s_{\max} = 0.5$ (gradient corresponding to $\approx 27^\circ$, conservatively below the $\mu = 0.8$ friction limit), and $r_{\max} = 0.03$ m. The gradient is computed via central finite differences and $\mathrm{std}(\cdot)$ is evaluated over a 5-point cross-shaped neighbourhood $\mathscr{N}$. A score of $\sigma = 0$ denotes ideal flat ground; $\sigma = 1$ denotes untraversable terrain.

3) Obstacle Cost Map: Cells with clearance $c \geq c_{\mathrm{lethal}} = 0.12$ m (the robot's step-over limit given a swing apex of 0.15 m) are classified as obstacles and inflated with a linearly decaying radial kernel of radius $r_{\mathrm{inf}} = 0.50$ m:

$$\mathscr{C}_{\mathrm{inf}}(d) = \mathscr{C}_{\mathrm{raw}} \cdot \max\left(0,\ 1 - \frac{d}{r_{\mathrm{inf}}}\right) \tag{2}$$

The inflation radius accounts for the robot's half-width plus a dynamic tracking error margin.

C. Terrain-Aware Global Planning

1) Weighted A Formulation:* We compute global waypoints via weighted A* on the 8-connected cost grid $\mathscr{G}$. Let $\mathscr{C}(n)$ denote the obstacle costmap value and $\sigma(n)$ the steppability score at node n. The edge cost from node n to neighbour n' is:

$$g(n') = g(n) + \|\mathbf{p}_{n'} - \mathbf{p}_n\| \cdot \left(1 + w_{\mathrm{obs}}\left(e^{\beta\,\mathscr{C}(n')} - 1\right) + w_{\mathrm{terr}} \cdot \sigma(n')\right) \tag{3}$$

with $w_{\mathrm{obs}} = 100$, $\beta = 4$, $w_{\mathrm{terr}} = 5$. The **exponential obstacle penalty** creates a steep cost gradient that forces paths well into free space. At 50% inflation ($\mathscr{C} = 0.5$), the multiplicative cost factor reaches $\approx 640\times$, effectively repelling paths from the inflation zone. The heuristic uses a $1.5\times$ Euclidean weight for faster search:

$$f(n) = g(n) + 1.5 \cdot \|\mathbf{p}_n - \mathbf{p}_{\mathrm{goal}}\| \tag{4}$$

2) Path Hysteresis: To prevent oscillatory replanning around symmetric obstacle configurations, cells belonging to the previous path $\mathscr{P}_{t-1}$ receive a discounted traversal cost:

$$g_{\text{hyst}}(n') = \begin{cases} g(n') & \text{if } n' \notin \mathscr{P}_{t-1} \\ 0.5 \cdot g(n') & \text{if } n' \in \mathscr{P}_{t-1} \end{cases} \tag{5}$$

The 50% discount prevents topological oscillation (e.g., flipping between passing left vs. right of a tree) while permitting route changes if a newly discovered obstacle blocks the path. The raw path is then smoothed via constrained Laplacian smoothing to round corners without intersecting obstacles.

D. Foothold-Quality-Aware MPPI

1) MPPI Preliminaries: We adopt the sampling-based MPPI framework [5]. Define the state $\mathbf{s}_t = [x_t, y_t, \psi_t, v_{x,t}, v_{y,t}, \omega_{z,t}]^T \in \mathbb{R}^6$ and control $\mathbf{u}_t = [v_{x,\text{cmd}}, v_{y,\text{cmd}}, \omega_{z,\text{cmd}}]^T \in \mathscr{U}$ with bounds:

$$\mathscr{U} = \{v_x \in [-0.25, 0.60],\ v_y \in [-0.5, 0.5],\ \omega_z \in [-1.75, 1.75]\} \tag{6}$$

The asymmetric v_x bounds reflect gait stability constraints on the quadruped.

2) Rollout Dynamics and Noise: We propagate each rollout using a first-order velocity lag model that captures the response latency of the underlying centroidal MPC:

$$v_{x,t+1} = v_{x,t} + \frac{(v_{x,\text{cmd}} - v_{x,t})\,\Delta t}{\tau}, \qquad \tau = 0.4\,\text{s} \tag{7}$$

with analogous updates for v_y and ω_z. Position integration follows the standard unicycle model over x, y, and ψ. Candidate control sequences $\mathbf{U}_i$ are generated by perturbing the nominal sequence with temporally correlated AR(1) noise:

$$\varepsilon_{i,t} = \rho\,\varepsilon_{i,t-1} + (1-\rho)\,\eta_{i,t}, \qquad \eta_{i,t} \sim \mathscr{N}(\mathbf{0}, \Sigma) \tag{8}$$

with $\rho = 0.5$. Correlated noise produces feasible, arc-like trajectories rather than kinematically unrealisable zig-zag paths.

3) Multi-Critic Cost Function: Each candidate trajectory i incurs a total cost C_i that is the sum of eight critic terms. We detail the two novel critics below, followed by a summary of all terms.

- **Turn-in-Place Critic (Crab-Walk Suppression):** When correcting path-following errors, quadrupeds commanded via body-frame velocity often exhibit simultaneous forward and lateral motion ("crab-walking"), which reduces the lateral stability margin. We introduce a critic that penalises forward velocity when the heading error $|\Delta\psi_t^{(i)}|$ is large:

$$C_{\text{turn}}^{(i)} = w_{\text{turn}} \frac{1}{H} \sum_{t=1}^{H} [\max(0, v_{x,t}^{(i)})]^2 \left[\max\left(0,\ |\Delta\psi_t^{(i)}| - \psi_{\text{dead}}\right)\right]^2 \tag{9}$$

  where $\psi_{\text{dead}} = 0.2$ rad $(= 11.46°)$. The quadratic coupling ensures the penalty is dominant for large heading errors. With $w_{\text{turn}} = 500$, the optimiser drops $v_x \to 0$ and uses ω_z to align the heading before moving forward.

- **Foothold-Quality-Aware (FQA) Critic:** Standard MPPI [5] evaluates trajectories by querying the costmap at the Center of Mass (COM) position. Because legged robots have distinct ground contact patches, we accurately predict where each foot will contact the ground during a rollout, shifting the terrain evaluation from discrete footstep planners [16], [17], [18] directly into the continuous trajectory rollout.
  Let $\mathbf{h}^l = (h_x^l, h_y^l)^T$ denote the body-frame hip offset for leg $l \in \{1, \ldots, 4\}$. The predicted world-frame foothold is:

$$\mathbf{f}^{l,t} = \begin{bmatrix} x^t \\ y^t \end{bmatrix} + \mathbf{R}_z(\psi^t)\,\mathbf{h}^l = \begin{bmatrix} x^t + h_x^l \cos\psi^t - h_y^l \sin\psi^t \\ y^t + h_x^l \sin\psi^t + h_y^l \cos\psi^t \end{bmatrix} \tag{10}$$

  Note: this assumes zero capture-point displacement, i.e., feet land directly below the hip. This is a valid approximation at low forward speeds where the Raibert velocity-dependent offset $\mathbf{v} \cdot T_{\text{stance}}/2$ is small relative to the hip spread.
  The terrain steppability σ from (1) is queried at each predicted foothold, and the FQA cost averages the squared penalty over all horizon steps H:

$$C_{\text{FQA}}^{(i)} = w_{\text{fqa}} \cdot \frac{1}{4H} \sum_{l=1}^{4} \sum_{t=1}^{H} \left[\sigma\left(\mathbf{f}^{l,t}\right)\right]^2 \tag{11}$$

  The squared penalty cleanly amplifies the distinction between mediocre and severely poor footholds, guaranteeing that MPPI avoids paths requiring steps across untraversable terrain.

The total cost aggregates standard Nav2 terms (Path-Follow, Path-Progress, Obstacle Clearance) alongside FQA ($w_{\text{fqa}} = 3.0$) and Turn-in-Place. Table I summarizes all terms.

TABLE I
COMPLETE MPPI CRITIC FORMULATION

Critic	**Weight** w	**Cost Definition**
Path-follow	80	Mean cross-track distance to A* path
Path-angle	120	Mean $\lvert\psi_t - \operatorname{atan2}(\Delta p_y, \Delta p_x)\rvert$
Turn-in-place	500	Eq. (9): $v_x^2 \cdot \max(0, \lvert\Delta\psi\rvert - 0.2)^2$
Path-progress	20	$(L - s_{\max})/L$; penalises lack of advance
Obstacle	30–300	Graduated: 300 (lethal), 150$\mathscr{C}$ (mod.), 30$\mathscr{C}$ (mild)
Slope	2.5	Mean $\lVert\nabla h_g\rVert$ along trajectory
FQA	**3.0**	**Eq. (11): Squared steppability at feet**
Smoothness	0.1	Mean $\lVert\Delta\mathbf{u}_t\rVert$ (control rate penalty)

4) Optimal Control via Softmax Weighting: The optimal control sequence $\mathbf{U}^*$ is computed as the information-theoretic weighted average:

$$\mathbf{U}^* = \sum_{i=1}^{K} w_i\,\mathbf{U}_i, \qquad w_i \propto \exp\left(-\frac{C_i - C_{\min}}{\lambda}\right) \tag{12}$$

with temperature $\lambda = 8.0$. After $N_{\text{iter}} = 5$ optimization iterations, the first step of the control sequence is extracted, rate-limited, and passed to the low-level controller.

E. Centroidal MPC and Low-Level Control

1) QP Formulation: The body-frame velocity commands $(v_x^*, v_y^*, \omega_z^*)$ from MPPI are passed to the low-level controller as tracking references. We model the quadruped using Single Rigid Body (SRB) dynamics and optimize ground reaction forces (GRFs) across a 16-step horizon (equivalent to one full 0.33 s trot cycle). This is formulated as a sparse Quadratic Program (QP):

$$\min_{\mathbf{X},\mathbf{U}} \sum_{k=0}^{N-1} \left[(\mathbf{x}_k - \mathbf{x}_k^{\text{ref}})^T \mathbf{Q} (\mathbf{x}_k - \mathbf{x}_k^{\text{ref}}) + \mathbf{u}_k^T \mathbf{R} \mathbf{u}_k \right] \tag{13}$$

Here, the state vector $\mathbf{x}_k$ tracks the robot's base position, orientation, linear, and angular velocities. The control vector $\mathbf{u}_k$ comprises the 3D forces at each stance foot. To ensure gait stability, the tracking error cost $\mathbf{Q}$ heavily penalizes deviations in body height ($q_z = 50$) and pitch/roll ($q_\theta = 20, q_\phi = 10$), while strictly lateral x, y position tracking is left loose ($q_x = q_y = 1$) since the MPPI layer handles spatial path guidance. The $\mathbf{R}$ matrix serves purely as a Hessian regularizer ($\mathbf{R} = 10^{-5}\mathbf{I}$).

2) Terrain-Adaptive Constraints: Crucially, the QP solver enforces friction pyramid limits ($\mu = 0.8$) to ensure the feet do not slip. On sloping terrain, friction cones aligned to world-gravity become physically invalid. To counter this, the contact surface normal $\hat{\mathbf{n}}$ at each leg's touchdown location is queried from the height map; the QP solver dynamically rotates the friction constraints into the terrain-local tangent plane.

3) MPPI–MPC Synergy: The FQA critic acts as a spatial pre-filter. Because MPPI actively penalizes center-of-mass trajectories that place peripheral legs into deep terrain variations, the reference velocities passed to the MPC natively guide the robot through clustered safe zones. This synergy drastically simplifies the low-level Raibert-based foothold selection planner, as it operates exclusively within bounds established as traversable by the hierarchical planner.

IV. Experimental Results

The robot traverses 5m from (-2,0) to (3,0) across 5 LiDAR-derived terrains from British Columbia forests, each with randomized obstacles: 15 tree trunks (cylinders), 3 rocks (spheres), and 2 fallen logs (capsules) — placed to block the direct path and force the planner to find detours. Positions are rejection-sampled to keep start/goal areas clear. The simulation duration is 20 secs on a randomized terrain.

Traversal time reports how long the robot took to reach within 0.75m of the goal. Body contacts counts collisions between the robot's body (excluding feet) and obstacles, indicating whether the robot physically struck any object or fell during navigation. Roll (deg) measures the mean absolute lateral body tilt during traversal, reflecting locomotion stability over uneven terrain. This tests generalization of the convex MPC + MPPI planner across varied terrain and obstacle layouts. (Fig 3) Table II reports traversal across five distinct LiDAR-derived British Columbia forest terrains, each with obstacle layouts. The full system reaches the goal

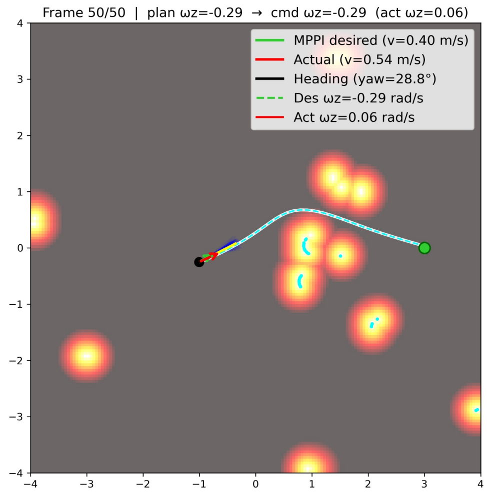

Fig. 3. Example frame from the MPPI-based local planner operating in a cluttered environment. The planner generates a smooth trajectory (cyan dashed line) toward the goal (green marker) while avoiding obstacles represented by the costmap heatmap. The green arrow denotes the desired velocity from MPPI, the red arrow shows the robot's actual velocity, and the black arrow indicates the current heading. The multiple dark blue lines near the robot depicts the MPPI rollouts. The planner commands an angular velocity of 0.29 rad/s, while the measured yaw rate is 0.06 rad/s.

in all five environments with zero body contacts, despite substantial variation in terrain geometry. Traversal times range from 12.5 s to 18.0 s, reflecting the differing detour lengths required to route around obstacle clusters, while mean roll remains bounded between 7.6° and 12.5° across all runs. The consistency of fall-free traversal across geometrically diverse terrains indicates that the foothold-quality-aware planner generalizes beyond a single hand-tuned scene, rather than overfitting to a specific obstacle configuration.

Environment	Time (s)	Roll (deg)	Contacts
bc_094p006_2_2_3_xli1m_utm10	14.7	12.5	0
bc_083e021_xli1m_utm11_2019	12.5	9.19	0
bc_093f077_xli1m_utm10_2019	18.0	9.62	0
bc_093f097_xli1m_utm10_2019	16.9	8.79	0
bc_093g045_xli1m_utm10_2019	16.2	7.63	0

TABLE II

Results across Five British Columbia LiDAR-derived terrain environments.[19]

V. Ablation Study

To systematically evaluate the contribution of each proposed component within our hierarchical navigation framework, we conducted a comprehensive ablation study. The experiments involve navigating a simulated Go2 quadruped over a challenging 5-meter unstructured terrain course (from $x = -2$ m to $x = 3$ m) with obstacles on a single hand-tuned scene. Each trial has a maximum duration of 18 seconds. The primary metrics for evaluation are: success rate (Goal Reached), final distance to the goal, Roll/Pitch Root Mean Square (RMS) to quantify stability, and the total number of body contacts (falls).

TABLE III

ABLATION STUDY RESULTS: 5M TRAVERSAL OVER UNEVEN TERRAIN

Configuration	Goal Reached	Final Dist. (m)	Roll RMS (rad)	Pitch RMS (rad)	Body Contacts
Case 0: Full System (Baseline)	**Yes (14.7s)**	**0.23**	**0.27**	**0.07**	**0**
Case 1: No Path Planning	No	2.65	0.12	0.05	0
Case 2: No FQA Steppability Critic	No	2.54	0.89	0.13	4
Case 3: No Terrain Friction Cones	Yes (15.8s)	0.42	0.36	0.18	6
Case 4: No Terrain COM Adaptation	Yes (17.9s)	0.72	0.73	0.16	8
Case 5: No Capture-Point Stepping	No	2.46	0.87	0.28	12
Case 6: No Terrain Foothold Selection	No	1.23	0.72	0.11	7
Case 7: Blind Locomotion	No	1.76	1.83	0.17	9

The results of the ablation study are summarized in Table III. We compare the full proposed system (Baseline) against seven degraded configurations, each with a specific feature disabled.

A. Analysis of Ablations

Case 0: Full System (Baseline). The complete framework successfully reaches the goal in 14.7 seconds with a final distance of 0.23 m. It maintains a low Roll RMS (0.27 rad, or 15.5°) and, crucially, achieves zero body contacts. This is the only configuration that demonstrates clean, stable traversal without any falls, establishing the baseline performance of the integrated system.

Case 1: No Path Planning. Disabling the MPPI planner forces the robot to rely on a naive goal-seeking heuristic (direct heading). While the robot remains stable (Roll RMS of 0.12 rad, zero falls), it gets stuck behind obstacles, failing to reach the goal (2.65 m remaining). The total path length is only 4.16 m compared to the baseline's 8.34 m. This demonstrates that while local stability is maintained by the low-level controller, high-level path planning is essential for navigating around complex obstacles and avoiding dead ends.

Case 2: No FQA Steppability Critic. Removing the Foothold-Quality-Aware (FQA) critic from the MPPI cost function allows the planner to select paths traversing geometrically feasible but dynamically unstable terrain (e.g., steep slopes or sharp edges). The robot fails to reach the goal, Roll RMS spikes to 0.89 rad (max roll 180°), and the robot suffers 4 body contacts.

This case corresponds to the standard MPPI local-planner configuration used in wheeled-robot navigation stacks, where trajectory costs are evaluated against the COM footprint without foothold-quality reasoning. Its failure to traverse the course (4 body contacts, 0.89 rad roll RMS) directly demonstrates the inadequacy of COM-centric critics for legged platforms and highlights the critical role of the FQA critic in preventing the planner from guiding the robot into unsteppable terrain regions — the gap motivating this work.

Case 3: No Terrain Friction Cones. When the MPC assumes flat-ground friction cones rather than adapting to the local surface normal, the robot reaches the goal but suffers large performance degradation (15.8 seconds, 0.42 m final distance). Roll RMS increases to 0.36 rad, and the robot incurs 6 body contacts. The MPC commands forces that violate actual friction limits on slopes, causing slippage and partial falls. Terrain-aware friction cones are thus vital for force feasibility and reliable slope climbing.

Case 4: No Terrain COM Adaptation. Disabling both terrain-aware COM height and orientation adjustments forces the robot to maintain a level body posture regardless of the underlying slope. The robot reaches the goal, but does so slowly (17.9 seconds) and with the worst performance among successful cases. Roll RMS reaches 0.73 rad, max roll hits 180°, and the robot suffers 8 body contacts. Fighting the terrain creates large moment arms and forces legs to their kinematic limits, proving that COM adaptation is crucial for maintaining workspace limits and balance.

Case 5: No Capture-Point Stepping. Disabling capture-point stepping (lateral foot widening during roll) and adaptive stability weighting removes the robot's primary reactive balance mechanism against lateral disturbances. The robot fails to reach the goal, with a Roll RMS of 0.87 rad and a total of 12 body contacts across 72 steps—the highest fall count of any ablation. Without capture-point stepping to catch the shifting Center of Mass, any disturbance quickly cascades into a catastrophic fall, making this the single most important feature for maintaining reactive balance on rough terrain.

Case 6: No Terrain Foothold Selection. Disabling both terrain-aware foothold selection (evaluating candidates based on slope, stability, and planarity) and the MPPI steppability critic forces the robot to step at nominal Raibert heuristic positions. The robot fails to reach the goal (1.23 m remaining), suffering 7 body contacts with a Roll RMS of 0.72 rad. Although the robot makes significant forward progress, indiscriminate foot placement on edges and unstable patches causes repeated stumbling.

Case 7: Blind Locomotion. Disabling all terrain-aware features—leaving only LiDAR-based obstacle detection—results in near-total failure. The robot fails to reach the goal, experiencing continuous tumbling with a Roll RMS of 1.83 rad (mean 92°) and 9 body contacts. The simulation terminates early due to continuous failure to recover. This catastrophic degradation validates the necessity of the proposed hierarchical approach: terrain perception and reactive adaptation are not optional enhancements, but fundamental requirements for legged locomotion in unstructured environments.

B. Summary of Contributions

The ablation results reveal a clear functional hierarchy within the proposed framework. Reactive balance mechanisms (capture-point stepping, foothold selection) are the primary defense against falls. Perception-informed path planning (MPPI, FQA critic) ensures the robot chooses dynamically viable routes, avoiding geometrically feasible but unstable terrain. Finally, terrain-aware centroidal dynamics (friction cones, COM adaptation) optimize force distribution and maintain kinematic reachability. Removing any single layer significantly degrades performance, and removing all layers (Case 7) results in total failure, affirming the necessity of the integrated system design.

VI. Conclusion and Future Work

In this paper, we presented a hierarchical navigation architecture for quadruped robots that explicitly bridges the gap between high-level path planning and low-level reactive locomotion. By integrating a Foothold-Quality-Aware (FQA) critic and a Turn-in-Place critic into a Model Predictive Path Integral (MPPI) local planner, the framework ensures that generated velocity commands adhere to the terrain's steppability constraints and preserve the robot's lateral stability margin. Coupled with a terrain-aware centroidal Model Predictive Controller (MPC) and reactive foothold adaptation mechanisms, our approach significantly reduces the frequency of catastrophic falls on unstructured terrain. Extensive ablation studies in simulated dense forest environments validated that removing any component of this perception-to-control pipeline drastically degrades locomotor performance, demonstrating that intelligent path planning and reactive balance are fundamentally codependent for robust legged navigation.

Despite the promising results demonstrated in simulations, several avenues remain open for future research. First, we aim to conduct more robust testing across a wider variety of extreme environments, including but not limited to stairs, sheer drop-offs, cliffs, and dynamic moving obstacles, to further evaluate the framework's operational limits.

Second, we plan to deeply integrate exteroceptive vision (e.g., RGB or semantic cameras) directly into the costmap generation process. This will enable semantic reasoning about physical surface properties—such as estimating terrain friction coefficients before stepping, or identifying non-geometric hazards like water, ice, and mud—which LiDAR-based elevation maps inherently cannot detect.

Thirdly, incorporating adaptive gait scheduling based on the navigation difficulty of the terrain and the desired traversal speed represents a crucial next step. For example, the system could autonomously transition to a highly stable, albeit slower, crawl gait when navigating treacherous, high-cost regions, while automatically shifting to a dynamic trot to maximize speed and efficiency across flatter ground.

Most critically, we plan to deploy the framework on a physical Unitree Go2. A successful sim-to-real transfer will primarily require integrating a robust onboard localization pipeline (e.g., LiDAR-inertial odometry), as the current system relies on ground-truth pose available only in simulation, along with careful compensation for actuator delays and sensor noise.

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